Numerical Analysis I

Quiz 9A Name:_____

Question 1.

The number of users of an online platform has grown rapidly over several years. The table below gives the number of users (in millions) at selected times.

Year	t	Users (millions), y
2015	0	2.5
2016	1	3.8
2017	2	5.6
2018	3	8.7
2019	4	12.9
2020	5	19.8
2021	6	29.5

Fit the exponential model

$$y = c_1 e^{c_2 t}$$

to the data using model linearization.

(a) Linearize the model by taking the natural logarithm of both sides. Express the resulting problem as a linear least squares give the coefficient matrix A and the right hand side b.

(b) Use the method of least squares to determine the coefficients a and b of the linearized model.

(c) Determine the parameters c_1 and c_2 of the original exponential model.

(d) Use the fitted model to predict the number of users in 2023.

(e) Explain why minimizing the sum of squared residuals for the linearized model is not equivalent to minimizing the sum of squared residuals for the original exponential model.

(a) Taking the natural logarithm of both sides of

$$y = c_1 e^{c_2 t},$$

we obtain

$$\ln(y) = \ln(c_1) + c_2 t.$$

Setting $x_1 = \ln(c_1)$ and $x_2 = c_2$, we obtain the linear model

$$\ln(y) = x_1 + x_2 t.$$

(b) The linear least squares problem is

$$\min_{x \in \mathbb{R}^2} \|Ax - b\|_2,$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} \ln(2.5) \\ \ln(3.8) \\ \ln(5.6) \\ \ln(8.7) \\ \ln(12.9) \\ \ln(19.8) \\ \ln(29.5) \end{bmatrix}.$$

(c) Since $x_1 = \ln(c_1)$ and $x_2 = c_2$, we have

$$c_1 = e^{x_1} \approx e^{0.916} \approx 2.50, \qquad c_2 = x_2 \approx 0.412.$$

Thus, the fitted exponential model is

$$\boxed{y \approx 2.50e^{0.412t}.}$$

Question 2.

Find the least squares solution of

$$\min_{x \in \mathbb{R}^2} \|Ax - b\|_2,$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 2 \\ 5 \\ 5 \end{bmatrix}.$$

The normal equations are

$$A^T Ax = A^T b.$$

We compute

$$A^T A = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 14 \\ 27 \end{bmatrix}.$$

Thus, we need to solve

$$\begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 14 \\ 27 \end{bmatrix}.$$

The least squares solution is

$$\boxed{x = \begin{bmatrix} \frac{17}{10} \\ \frac{6}{5} \end{bmatrix}.}$$